

4.4.1 One surgical step in a multi-step method

Opinion G 1/04 (OJ 2006, 334) made it clear that a method claim falls under the prohibition of Art. 52(4) EPC 1973 if it includes at least one feature defining a physical activity or action that constitutes a method step for treatment of the human or animal body by surgery or therapy.

In G 1/07 the Enlarged Board upheld this principle confirmed in opinion G 1/04 (OJ 2006, 334). This principle is not only formally justified by the fact that the exclusion under Art. 53(c) EPC does not contain any limitation as to the defined methods being excluded only when claimed as such. More importantly, it is also justified as to substance, i.e. it enables the legislative purpose served by the exclusion to be achieved (see in this chapter I.B.4.1.).

4.4.2 Treatment by surgery not limited to surgery for a therapeutic purpose

In G 1/07 (OJ 2011, 134) the Enlarged Board of Appeal concluded that neither the legal history nor the object and purpose of the exclusions from patentability in Art. 53(c) EPC would justify a limitation of the term "treatment by surgery" to curative surgery (overruling T 383/03). Such a limitation would be contrary to the ordinary understanding of the word "surgery" as referring to the nature of the treatment rather than its purpose, and contrary to the fact that Art. 53(c) EPC defines three separate alternative exclusions thereby suggesting that these are not merely identical in scope. The Enlarged Board of Appeal observed in G 1/07 that the comparison between T 383/03 and T 1172/03 showed how inconsistent the decisions to be made could become if the term "treatment by surgery" was seen as limited to therapeutic surgery only. The term "treatment by surgery" is therefore not to be interpreted as being confined to surgical methods pursuing a therapeutic purpose (G 1/07). As stated in the Guidelines (Guidelines G-II, 4.2.1.1 – March 2022 version), the term "surgery" thus defines the nature of the treatment rather than the purpose.

4.4.3 Towards a new concept of "treatment by surgery"

In G 1/07 the Enlarged Board of Appeal disagreed with the EPO's practice of broadly interpreting the surgical character of interventions.

A narrower understanding of "treatment by surgery" was required. Any definition of the term "treatment by surgery" must cover the kind of interventions which represent the core of the medical profession's activities, i.e. the kind of interventions for which their members are specifically trained and for which they assume particular responsibility. Such a narrower understanding rules out uncritical methods involving only a minor invention and no substantial health risks.

It is for the departments of first instance and the boards to define the boundaries of a more narrowly construed concept of "treatment by surgery", based on the technical reality of the individual case under consideration.

The required medical expertise and the health risk involved may not be the only criteria which may be used to determine that a claimed method actually is a "treatment by surgery" within the meaning of Art. 53(c) EPC. It appears that what is to be understood by "surgery" in the medical sense is to a large extent a matter of convention. Thus, in order to be surgical, it is not necessary that the intervention be invasive or that tissues be penetrated (T. 5/04). The scope of what is surgery may change with time and with new technical developments emerging, as was already acknowledged in decision T. 182/90.

In G. 1/07, with regard to the facts of the case, the Enlarged Board decided that a claimed imaging method, in which, when carried out, maintaining the life and health of the subject is important and which comprises or encompasses an invasive step representing a substantial physical intervention on the body which requires professional medical expertise to be carried out and which entails a substantial health risk even when carried out with the required professional care and expertise, is excluded from patentability as a method for treatment of the human or animal body by surgery pursuant to Art. 53(c) EPC.

4.4.4 "Treatment by surgery" in the case law since G 1/07

a) Surgical nature of a method step

In T. 992/03 of 4 November 2010, ¹²⁹Xe gas was used as part of the claimed method. The board referred to G. 1/07 according to which "there is an exclusion from patentability as a surgical method only if the health risk is associated with the mode of administration and not solely with the agent as such" and any anaesthetic effect of the ¹²⁹Xe gas thus did not fall under the exclusion.

b) Assessment of "substantial health risks"

In T. 663/02 the board held that the step of "injecting the magnetic resonance contrast agent into a vein remote from the artery" may be considered as representing a minor routine intervention which did not imply a substantial health risk when carried out with the required care and skill. Such acts would be ruled out from the scope of the application of the exclusion clause pursuant to Art. 53(c) EPC following the narrow understanding advocated by the Enlarged Board of Appeal (G. 1/04 and G. 1/07). A possible way of assessing health risks is to use a risk matrix permitting the levels of likelihood and health impact of a complication of a medical act with regard to a large number of patients to be combined, so as to obtain statistical health risk scores which may be used to decide what action should be taken.

However, in T. 1075/06 the board held that venipuncture of blood donors and the extraction of blood from a donor's body represent substantial physical interventions on the body which require professional medical expertise to be carried out and which entail a substantial health risk even when carried out with the required professional care and expertise. A method claim comprising steps encompassing such procedures is a method for treatment of the human body by surgery.

Similarly, in T.1695/07, the board held that a blood manipulation process involving the continuous removal of blood from a patient, its subsequent flowing through a circulating line of an extracorporeal circuit and its re-delivery to the patient was a method of treatment of the human body by surgery. It stated that even when the process was carried out with the required medical professional care and expertise, it involved "substantial health risks" for the patient. A health risk was considered to qualify as "substantial" whenever it went beyond the side effects associated with treatments such as tattooing, piercing, hair removal by optical radiation, micro abrasion of the skin as mentioned in G.1/07. A factual analysis of absolute or relative risks and their likelihood of occurrence based on objective evidence was hardly feasible and should therefore not be required.

In T.2699/17 the application related to the guided expansion of an elastomeric material within the sulcus of a tooth. In this way, the gingiva was retracted from the tooth, such that an appropriate impression of the tooth could be obtained, which was then used in the manufacture of the crown restoration. The criteria as defined in G.1/07 were applied and the board found that minor injury of the epithelium could occur. It then had to examine whether the method qualified as "substantial physical intervention on the body", i.e. whether the health risk was a substantial health risk within the meaning of G.1/07. For that evaluation, different approaches had been suggested in the case law, namely, the 'risk matrix' in decision T.663/02 and a 'more abstract risk criterion' in decision T.1695/07. The board followed the latter approach, which was limited to the question "Is a certain health risk present?" and "Is it substantial?" The board held that the risks here were at a level equal to those present in the methods which G.1/07 considered not to involve a substantial health risk (see also T.467/18).

In T.434/15 the inventions concerned apheresis, a method in which blood was removed from a person, passed through an apparatus for separating and collecting a particular constituent of the blood (in the present case, stem/progenitor cells) and re-transfused without the collected constituent. The fact that a certain method was routine for a highly specialised centre could not, according to the board, automatically lead to the conclusion that such a method would generally be a safe and routine technique. Apheresis was held to be a method using invasive techniques that allowed for extra corporeal manipulation of an organ of the human body. There were considerable health risks involved. Furthermore, apheresis could not be considered to be a generally safe, simple, routine procedure. Consequently, apheresis had to be seen as a method of surgery within the meaning of Art. 53(c) EPC.

c) Surgical step part of claimed method or simply a preparatory measure?

The boards have had to consider whether particular surgical steps form part of the claimed method or are simply preparatory measures which cannot be deemed such a part.

In T.992/03 of 4 November 2010, methods were claimed for MR imaging the pulmonary and/or cardiac vasculature of a subject, using dissolved-phase polarised ¹²⁹Xe gas. The board stated that delivering polarised ¹²⁹Xe gas to the subject and initialising the MR system were preparatory steps which, however, did not form part of the contribution of the invention to the art. The method claims thus did not comprise "an invasive step

representing a substantial physical intervention on the body which requires professional medical expertise to be carried out and which entails a substantial health risk even when carried out with the required professional care" (G 1/07).

In T 2438/11 the claim was directed to a method for instrument alignment comprising steps whereby a surgical drill or mill was aligned with a target point. Said alignment was to take place in the patient's body, in direct physical contact with the bone. The board held that the very act of alignment, i.e. actively moving an instrument of this kind within the patient's body, was a substantial physical intervention on the body and thus a method step for treatment of the human body by surgery. The claimed method for instrument alignment, which explicitly included the (active) alignment step, also differed fundamentally from a mere (passive) position-tracking method of the kind at issue in T 836/08 (see below).

d) Surgical step not in the claim but covered by it?

In T 836/08, claim 1 concerned a method for tracking the position of the distal end of a bone guide wire, using a medical optical tracking and navigation system. The board found that, while this meant that the reference apparatus had to be attached to, and the wire introduced into, the bone, these steps were not part of the expressly claimed method. The fact that the method was performed after, or even during, a surgical intervention on the body did not mean that the claimed position-tracking method as such was a method of treatment by surgery.

The board decided otherwise in T 923/08: where a method for determining measurements of the human or animal body necessarily entailed a surgical step consisting of fixing to the human or animal body a measuring device indispensable for performance of the method, that step had to be deemed an essential feature of the method which was covered by such a method, even if none of the claimed method features expressly related to this step. Such a method was excluded from patentability under Art. 53(c) EPC. Moreover, such a surgical step could not be removed from the scope of the claimed method by means of wording stating that the surgically placed measuring device was already fixed to the body before the start of the method, or by means of a disclaimer, as this would breach Art. 84 EPC (1973).

In T 2102/12 the claim related to medical robotic systems. The appellant sought to protect a method performed by a device for determining and displaying distance information, the claimed method being performed contemporaneously and in parallel with an unclaimed, surgical process. The board held that these two "types" of method steps were inextricably interrelated. The board found, that the presently claimed method of intertwined non-surgical ("claimed") and surgical ("unclaimed") steps was not comparable to that underlying T 836/08.

In T 1631/17 too, the wording of the claim did not explicitly mention any method step that could be considered surgical within the meaning of Art. 53(c) EPC; however, the claimed method included such a step. The surgical method step in question fell within the claimed method in terms of both when and where it was done. The claimed method could not be

performed without immediately performing this intermediate step, and this was sufficient for it to fall under the exception in Art. 53(c) EPC (G 1/07), OJ 2011, 134).

In T 699/12 the board noted that according to the jurisprudence of the boards there was no uniform approach how to interpret method claims, in which no particular therapeutic/surgical step was claimed, although it derived from the context that such a therapeutic/surgical step would be present in relation to the claimed method. In T 699/12, the claimed method implied an irradiation of the patient, because otherwise a quantification of the dose delivery in the radiotherapy treatment of the patient would not be possible. However, when carefully considering the wording of the claims, there was no basis for identifying a step like an "intermediate treatment of irradiating the patient for therapeutic purposes" that was de facto not claimed. Hence, claim 1 could not be interpreted to include a step of irradiating the patient.

e) The apparatus claim is not a disguised method claim

In T 1798/08, although all claims were directed to a visual prosthesis, i.e. an apparatus, they were objected to by the opponent under Art. 53(c) EPC as relating to a surgical method. The features "suitable to be located on the body of the user outside a wall of the sclera and attached to the sclera" and "suitable to be located implanted in the eye behind the iris", the opponent argued, related to a method for treatment of the human or animal body by surgery and thus transformed the claim into a "disguised" method claim, even though it was notionally directed to a device.

The board dismissed this objection. It observed that Art. 53(c), second sentence, EPC specified that the provision did not apply to products. The claim category "products" included apparatus. Accordingly, the provisions of Art. 53(c) EPC did not normally apply to apparatus claims. The fact that some features of the claimed apparatus were functionally defined in relation to the body of the patient did not, in itself, transform the apparatus claim into a method claim (T 712/93, T 1695/07; see also T 1407/08).

f) Unpatentable claims for products that can only be produced in a surgical step

Pursuant to the second clause of Art. 53(c) EPC, the exclusion from patentability "shall not apply to products, in particular substances or compositions, for use in any of these methods."

In G 1/07 it was found that a single surgical method step was enough to regard a multi-step method as being excluded from patentability. It was not a question of whether a medical practitioner would then infringe the claimed method. In decision T 775/97 this approach was transferred to particular products. The claim related to a new product which had been produced from two components, both known per se, by means of a surgical step in the human body. The board in T 775/97 held that no European patent could be granted with claims directed to a new and even possibly inventive way of using materials or devices, in particular endoprostheses, involving a treatment by surgery. This was held to be equally true for product claims defined by a construction which was only arrived at in the human or animal body following a surgical method step.